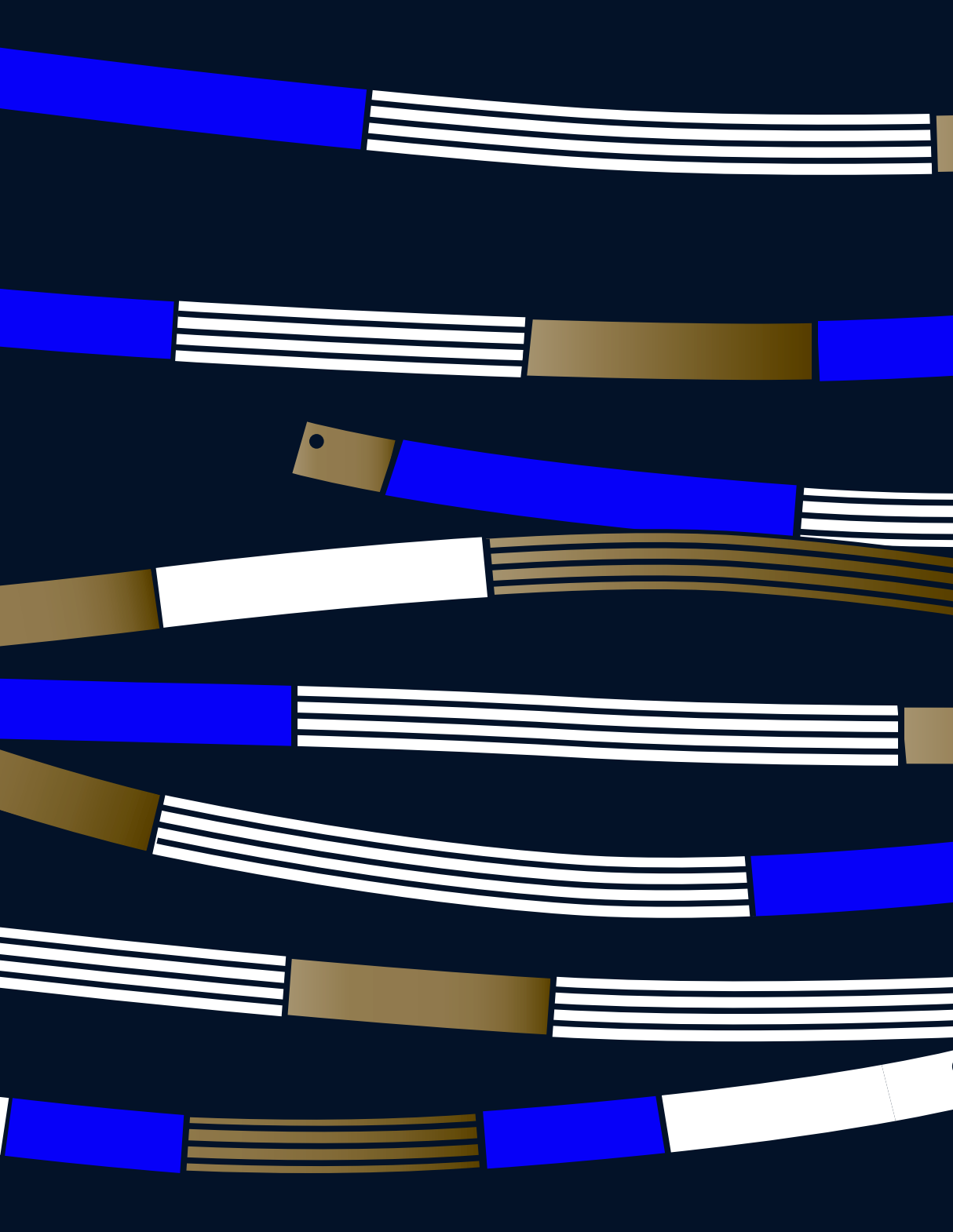
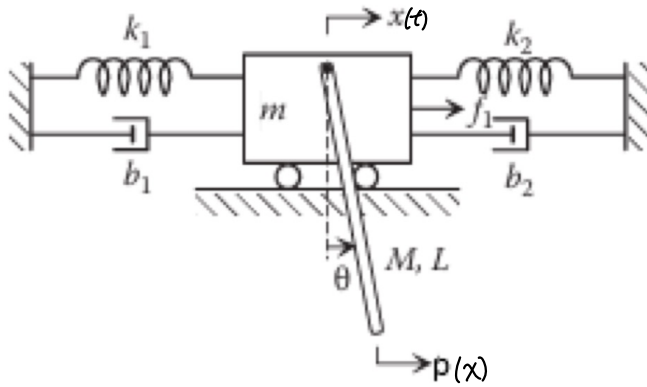






$F = \text{entrada}$

$x(t) = \text{Salida}$

Perturbación, $P(t) = A \cdot \text{Sen}(\omega t)$

Energía Cinética:

$$\frac{1}{2} m \dot{x}^2 + \frac{1}{2} \cdot M \cdot (V_{\text{péndulo}} + \dot{x} + \dot{p})^2$$

$$V_{\text{péndulo}} = \sqrt{\dot{x}_{\text{pen}}^2 + \dot{y}_{\text{pen}}^2}$$

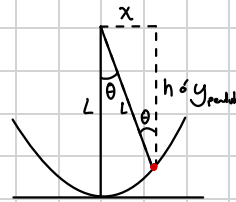
$$x_{\text{péndulo}} = L \cdot \text{Sen} \theta \longrightarrow \dot{x}_{\text{péndulo}} = L \cdot \text{Cos} \theta \cdot \dot{\theta}$$

derivando...

$$y_{\text{péndulo}} = -L \cdot \text{Cos} \theta \longrightarrow \dot{y}_{\text{péndulo}} = L \cdot \text{Sen} \theta \cdot \dot{\theta}$$

$$V_{\text{péndulo}} = \sqrt{(L \cdot \text{Cos} \theta \cdot \dot{\theta})^2 + (L \cdot \text{Sen} \theta \cdot \dot{\theta})^2} \longrightarrow L^2 \cdot \dot{\theta}^2 \cdot (\text{Sen}^2 \theta + \text{Cos}^2 \theta)$$

$$V_{\text{péndulo}} = \sqrt{L^2 \cdot \dot{\theta}^2} = L \cdot \dot{\theta}$$



$$\text{Cos} \theta = \frac{h}{L}$$

$$h = \text{Cos} \theta \cdot L$$

$$\text{Sen} \theta = \frac{x_{\text{pendulo}}}{L}$$

$$x_{\text{pendulo}} = L \cdot \text{Sen} \theta$$

Identidad Trigonométrica

$$K = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} \cdot M \cdot (L \cdot \dot{\theta} + \dot{x} + \dot{p})^2$$

Energía Potencial:

$$U_e = \frac{1}{2} \cdot K \cdot x^2 \longrightarrow E. \text{potencial elástica}$$

$$U_g = M \cdot g \cdot h \longrightarrow E. \text{potencial gravitacional}$$

$$U = \frac{1}{2} \cdot K_1 \cdot x^2 + \frac{1}{2} \cdot K_2 \cdot x^2 - M g L \text{Cos} \theta$$

$$U = \frac{1}{2} x^2 (K_1 + K_2) - M g L \text{Cos} \theta$$

$$\mathcal{L} = K - U$$

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} M (L \cdot \ddot{\theta} + \dot{x} + \dot{p})^2 - \frac{1}{2} x^2 (K_1 + K_2) + M g L \cos \theta$$

Coordenadas Generalizadas:

$$q_1 = x \rightarrow \dot{q}_1 = \dot{x}$$

$$q_2 = \theta \rightarrow \dot{q}_2 = \dot{\theta}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = \frac{F_{ext} + F_{nc}}{\tau_{ext} + \tau_{nc}}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} :$$

$$m \ddot{x} + M (L \ddot{\theta} + \ddot{x} + \ddot{p}) + x (K_1 + K_2) = F - (B_1 + B_2) \dot{x}$$

$$(m+M) \ddot{x} + M L \ddot{\theta} + M \ddot{p} + (B_1 + B_2) \dot{x} + (K_1 + K_2) x = F \quad (1)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} :$$

$$M (L \ddot{\theta} + \ddot{x} + \ddot{p}) + M g L \sin \theta = 0$$

$$L \ddot{\theta} + \ddot{x} + \ddot{p} + g L \sin \theta = 0 \quad (2)$$

Se despeja $\ddot{\theta}$ de (2) y se reemplaza en (1):

$$\ddot{\theta} = -\frac{\ddot{x}}{L} - \frac{\ddot{p}}{L} - \frac{g \sin \theta}{L}$$

$$(m+M) \ddot{x} - M \ddot{x} - M \ddot{p} - M g \sin \theta + M \ddot{p} + (B_1 + B_2) \dot{x} + (K_1 + K_2) x = F$$

$$m \ddot{x} - M g \sin \theta + (B_1 + B_2) \dot{x} + (K_1 + K_2) x = F$$

$$\ddot{x} = \frac{F + M \ddot{p} + M g \sin \theta - (B_1 + B_2) \dot{x} - (K_1 + K_2) x}{m}$$

Se reemplaza $\ddot{x}$ en ② y se despeja $\ddot{\theta}$:

$$\ddot{\theta} = -\frac{\ddot{x}}{L} - \frac{\ddot{p}}{L} - \frac{g \sin \theta}{L}$$

$$\ddot{\theta} = -\frac{F + Mg \sin \theta - (B_1 + B_2) \dot{x} - (K_1 + K_2) x}{mL} - \frac{\ddot{p}}{L} - \frac{g \sin \theta}{L}$$

$$\ddot{\theta} = \frac{-F - M\ddot{p} - Mg \sin \theta + (B_1 + B_2) \dot{x} + (K_1 + K_2) x - m\ddot{p} - mg \sin \theta}{mL}$$

$$\ddot{\theta} = \frac{-F - (M+m)\ddot{p} - (M+m)g \sin \theta + (B_1 + B_2) \dot{x} + (K_1 + K_2) x}{mL}$$

Puntos de Equilibrio ($\ddot{x} = \dot{x} = \ddot{\theta} = \dot{\theta} = \ddot{p} = 0$) $\sin \theta \approx \theta = 0$

$$(K_1 + K_2) x = F \quad F = (K_1 + K_2) x$$

Puntos de Operación: $\bar{x} = 0,5m$

$$\dot{x} = Ax + Bu + E\ddot{p} \quad y = Cx$$

$$\begin{bmatrix} \dot{x}_\delta \\ \dot{v}_\delta \\ \dot{\theta}_\delta \\ \dot{\omega}_\delta \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_1 + K_2}{m} & -\frac{B_1 + B_2}{m} & \frac{Mg}{m} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{K_1 + K_2}{mL} & \frac{B_1 + B_2}{mL} & -\frac{(M+m)g}{mL} & 0 \end{bmatrix}}_A \begin{bmatrix} x_\delta \\ v_\delta \\ \theta_\delta \\ \omega_\delta \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{mL} \end{bmatrix}}_B F_\delta + \underbrace{\begin{bmatrix} 0 \\ \frac{M}{m} \\ 0 \\ -\frac{M+m}{mL} \end{bmatrix}}_E \ddot{p}_\delta$$

$$y = [1 \ 0 \ 0 \ 0] x_\delta$$

$$m = 1 \text{ Kg} ; M = 2 \text{ Kg} ; L = 0,5m ; K_1 + K_2 = 10 \text{ N/m} ; g = 9,81 \text{ m/s}^2$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -10 & -1 & 19,62 & 0 \\ 0 & 0 & 0 & 1 \\ 20 & 2 & -58,86 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$$

$$Ff = \frac{s^2 + 19,62}{s^4 + s^3 + 68,86s^2 + 19,62s + 196,2} \quad n=4$$

Polos:

$$-0,13266 \pm 1,72328j \quad -0,36734 \pm 8,09585j$$

$s_{1,2}$ $s_{3,4}$

Dominantes rápidos y despreciables

$$(s + 0,1327 - 1,7233j)(s + 0,1327 + 1,7233j) = s^2 + 2 \cdot 0,1327s + 0,1327^2 + 1,7233^2 = s^2 + 0,2654s + 2,9856$$

$$\frac{19,62}{196,2} = 0,1$$

$$\frac{1}{2,9856} \approx 0,335$$

$$\frac{0,1}{0,335} \approx 0,2985$$

$$Ff = \frac{0,2985}{s^2 + 0,2654s + 2,9856} \quad n=2$$

← Función de transferencia reducida con polos Dominantes

Forma Canónica Controlable:

$$A = \begin{bmatrix} 0 & 1 \\ -2,9856 & -0,2654 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 0,2985 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 \end{bmatrix}$$

$$S = \begin{bmatrix} B & AB \end{bmatrix}$$

$$S = \begin{bmatrix} 0 & 1 \\ 1 & -0,2654 \end{bmatrix}$$

$$\det(s) = -1 \neq 0 \checkmark \quad \begin{matrix} \text{Es} \\ \text{Controlable} \end{matrix}$$

Forma Canónica Observable:

$$A = \begin{bmatrix} 0 & -2,9856 \\ 1 & -0,2654 \end{bmatrix}$$

$$B = \begin{bmatrix} 0,2985 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 \end{bmatrix}$$

$$\sigma = \begin{bmatrix} c \\ CA \end{bmatrix}$$

$$\sigma = \begin{bmatrix} 0 & 1 \\ 1 & -0,2654 \end{bmatrix}$$

$$\det(\sigma) = -1 \neq 0 \checkmark \quad \begin{matrix} \text{Es} \\ \text{Observable} \end{matrix}$$

$$Ff = \frac{0,2985}{s^2 + 0,2654s + 2,9856} \cdot \frac{1}{s} \quad \text{ess} = 0$$

$$t_{sla} = 10s$$

$$\gamma = 0,5$$

$$\omega_n = \frac{4}{\gamma \cdot t_{sla}} = 0,8$$

Controlador: PID

$$K_p + \frac{K_i}{s} + K_d s = \frac{K_p s + K_i + K_d s^2}{s}$$

$$0,2985 \cdot (K_p s + K_i + K_d s^2) + (s^2 + 0,2654s + 2,9856)s = \text{Pol.D}$$

$$s^3 + s^2(0,2985 K_d + 0,2654) + s(0,2985 K_p + 2,9856) + 0,2985 K_i = \text{Pol.D}$$

$$s^2 + 2\gamma\omega_n s + \omega_n^2 \rightarrow (s^2 + 0,8s + 0,64) \cdot (s + 4) = \text{Pol.D}$$

$$s^3 + s^2(4,8) + s(3,84) + 2,56 = \text{Pol.D}$$

$$s^3 \quad | = |$$

$$s^2 \quad 0,2985 K_d + 0,2654 = 4,8 \quad K_d = 15,19$$

$$s^1 \quad 0,2985 K_p + 2,9856 = 3,84 \quad K_p = 2,86$$

$$s^0 \quad 0,2985 K_i = 2,56 \quad K_i = 8,576$$

K para F.C.C $\rightarrow$ Ackerman

$$\hat{A} = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix} \quad \hat{B} = \begin{bmatrix} B \\ 0 \end{bmatrix} \quad \hat{A} = \begin{bmatrix} 0 & 1 & 0 \\ -2,9856 & -0,2654 & 0 \\ -0,2985 & 0 & 0 \end{bmatrix} \quad \hat{B} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\hat{s} = \begin{bmatrix} 0 & 1 & -0,2654 \\ 1 & -0,2654 & -2,9152 \\ 0 & 0 & -0,2985 \end{bmatrix}$$

$$\det(\hat{s}) = 0,2985 \neq 0$$

Ackerman ✓

$$K = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \cdot \hat{S}^{-1} \cdot \phi A$$

$$\hat{S}^{-1} = \begin{bmatrix} 0,2654 & 1 & -10 \\ 1 & 0 & -0,8891 \\ 0 & 0 & -3,3501 \end{bmatrix}$$

$$s^3 + s^2(4,8) + s(3,84) + 2,56 = \text{Pol.D}$$

$\alpha_1 \quad \alpha_2 \quad \alpha_3$

$$\phi A = \left[\hat{A}^3 + \alpha_1 \hat{A}^2 + \alpha_2 \hat{A} + \alpha_3 I \right]$$

$$\phi A = \begin{bmatrix} -10,9785 & -0,3491 & 0 \\ 1,0422 & -10,8859 & 0 \\ -0,255 & -1,3536 & 2,56 \end{bmatrix}$$

$$K = \begin{bmatrix} 0,8544 & 4,5346 & -8,5762 \end{bmatrix}$$

$K_1 \quad K_2 \quad K_i$

K para F.C.O $\rightarrow$ Ackerman

$$\hat{A} = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}$$

$$\hat{B} = \begin{bmatrix} B \\ 0 \end{bmatrix}$$

$$\hat{A} = \begin{bmatrix} 0 & -2,9856 & 0 \\ 1 & -0,2654 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\hat{B} = \begin{bmatrix} 0,2985 \\ 0 \\ 0 \end{bmatrix}$$